

ATEEB AHMED KHAN

Machine Learning Engineer · Computer Vision & LLM/RAG Systems

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SUMMARY

ML Engineer with hands-on experience building and deploying machine learning applications in **Computer Vision** and **LLM/RAG systems**. Experienced in developing end-to-end ML projects, from data preparation and model training to deployment and production inference. Built a **YOLO11**-based MRI tumor detector (95.1% mAP@0.5), a real-time face mask detection system using **DenseNet201** (98.99% accuracy), and a **RAG** application using **LangChain**, **ChromaDB**, and **LLaMA 3.3-70B**. Proficient in **Python**, **TensorFlow**, and **PyTorch**, and continuously expanding my expertise across **Artificial Intelligence**, **Machine Learning**, and **Generative AI**.

EXPERIENCE

DSIAR Tech

07/2025 – Present

AI/ML Mentor (Part-Time)

Remote

- Engineered a structured, production-oriented AI/ML curriculum spanning **25 structured modules comprising 100–125 video lessons** across completed AI and ML courses, with a Deep Learning track in active development. Each course culminating in a capstone project requiring learners to build and deploy a working ML application.
- Translated complex ML theory — including **model evaluation**, **feature engineering**, and **inference optimization** into modular, project-driven teaching units structured around hands-on implementation.

Labmentix

07/2025 – 12/2025

AI/ML Engineer (Intern)

Remote

- Developed and deployed end-to-end ML pipelines for data ingestion, preprocessing, model training, and production inference using modular, object-oriented Python components designed for reproducibility and reuse across projects.
- Deployed ML models as interactive web applications on **Google Colab + Streamlit**, integrating TensorFlow inference with a functional UI layer and implementing prediction logging to track output behavior post-deployment.
- Designed **reproducible training pipelines** with automated validation checkpoints, improving model reliability across iterative builds and eliminating manual evaluation steps.

Feynn Labs

09/2022 – 11/2022

Machine Learning Intern

Remote

- Developed ML-driven market segmentation models for EV industry analysis, applying **feature engineering** and clustering techniques to structure raw trend data into actionable product prototype inputs, directly informing two AI product scoping decisions.
- Built and evaluated end-to-end preprocessing and training pipelines in Python using **Scikit-learn** and **Pandas**, demonstrating proficiency in optimizing data transformation workflows for iterative model development.

PROJECTS

Document Q&A RAG System

[Live Demo ↗](#)

LangChain · ChromaDB · LLaMA 3.3-70B · Groq API · all-MiniLM-L6-v2 · PyMuPDF

- Built and deployed an end-to-end RAG pipeline** from PDF ingestion and chunking to semantic retrieval and LLM generation, deployed as a publicly accessible Streamlit application.
- Engineered a **modular 4-component system** (ingest, retriever, generator, UI) enabling independent testing and redeployment of each stage; reduced end-to-end query response time via async retrieval and Groq's inference API.
- Selected **all-MiniLM-L6-v2** over larger embedding models after benchmarking retrieval precision vs. latency trade-offs on domain-specific document corpora.

Brain Tumor Detection — YOLO11

[Live Demo ↗](#)

YOLO11 · PyTorch · Streamlit · Medical Imaging

- Designed and deployed a real-time MRI tumor detection system achieving **95.1% mAP@0.5** and 92.7% precision, with sensitivity-tuned thresholds to minimize false negatives in a clinical-risk context, prioritizing recall over precision by design.
- Built an end-to-end inference pipeline from MRI scan image input through YOLO11 object detection to annotated output rendering, deployed as a **low-latency web application** on Streamlit with sub-second inference per scan.

Real-Time Face Mask Detection

[Live Demo ↗](#)

DenseNet201 · TensorFlow Lite · Mediapipe · Streamlit

- Trained a **DenseNet201** classifier on a 992-image dataset achieving 98.99% accuracy; compressed and converted to **TensorFlow Lite** for edge-optimized inference, reducing model size and enabling deployment on resource-constrained environments.
- Integrated **Mediapipe** face detection as a preprocessing stage into a unified inference pipeline, enabling real-time video stream classification at 30fps with a modular, production-ready architecture.

SKILLS

Core Engineering

Python · SQL · OOP · REST APIs · Docker · Git · Flask

Generative AI & LLMs

LangChain · RAG pipelines · ChromaDB · LLaMA 3.3 · Groq API · Prompt Engineering

ML / Deep Learning

TensorFlow · PyTorch · Scikit-learn · Keras · YOLO

Computer Vision & Data

OpenCV · MediaPipe · NumPy · Pandas · Feature Engineering · Streamlit

EDUCATION

IU International University of Applied Sciences, Berlin

07/2023 – 06/2026

MSc Computer Science · Distance Learning · GPA 1.8 (German Scale)
Thesis: “Brain Tumor Detection Using Vision Transformers and XAI”

MIT World Peace University, Pune

2018 – 2022

B.Tech Robotics & Automation · GPA 8.99 / 10

Pune

CERTIFICATIONS

Machine Learning Specialization · Coursera / Stanford Online
Deep Learning & OpenCV for Computer Vision · Udemy

Google Data Analytics · Coursera / Google